Auteur: Stan Van Campenhout
Studierichting: Informatica
Oefeningenreeks 4A nr. 46.2

$$\int \frac{-3x+3}{2x^4+5x^2+2}dx = \int \frac{-3x+3}{(x^2+2)(2x^2+1)}dx = \frac{1}{2}\int \frac{-3x+3}{(x^2+2)\left(x^2+\frac{1}{2}\right)}dx$$

$$\to \frac{-3x+3}{(x^2+2)\left(x^2+\frac{1}{2}\right)} = \frac{Ax+B}{x^2+2} + \frac{Cx+D}{x^2+\frac{1}{2}}$$

$$A\left(\sqrt{2}i\right)+B = \frac{-3\sqrt{2}i+3}{-2+\frac{1}{2}} = -2-2\sqrt{2}i$$

$$\Rightarrow A = 2, B = -2$$

$$\Rightarrow C\left(\frac{\sqrt{2}}{2}i\right)+D = \frac{\frac{-3\sqrt{2}}{2}i+3}{-\frac{1}{2}+2} = 2-\sqrt{2}i$$

$$\Rightarrow C = -2, D = 2$$

$$= \int \frac{x-1}{x^2+2}dx - \int \frac{x-1}{x^2+\frac{1}{2}}dx$$

$$= \frac{1}{2}\int \frac{1}{x^2+2}d\left(x^2+2\right) - \int \frac{1}{x^2+2}dx - \frac{1}{2}\int \frac{1}{x^2+\frac{1}{2}}d\left(x^2+\frac{1}{2}\right) + \int \frac{1}{x^2+\frac{1}{2}}dx$$

$$= \frac{1}{2}\ln|x^2+2| - \frac{1}{\sqrt{2}}\operatorname{Bgtan}\left(\frac{x}{\sqrt{2}}\right) - \frac{1}{2}\ln\left|x^2+\frac{1}{2}\right| + \sqrt{2}\operatorname{Bgtan}\left(\sqrt{2}x\right) + c$$

References